



Girraween High School

2024 Year 11 Mathematics Extension 1 Yearly Examination September-2024

General Instructions

- Working Time - 120 minutes
- Reading Time 10 minutes
- Write using black or blue pen only
- Calculators approved by NESA may be used
- For questions in Section II, Show relevant Mathematical reasoning and/ or calculations

Total Marks: 90

Section I - 5 marks (pages 1 and 2)

- Attempt questions 1-5
- Allow about 10 minutes for this section

Section II - 85 marks (pages 3-9)

- Attempt questions 6-11,
- **Start each question on a new page**
- Allow about 110 minutes for this section

Section I

Multiple Choice Questions

Question 1 (1 mark)

What is the domain of the function $f(x) = \sqrt{x^2 - 2x}$

- A. $x \geq 2$
- B. $x > 2$
- C. $x \leq 0$ or $x \geq 2$
- D. $x < 0$ or $x > 2$

Question 2 (1 mark)

Given $f(x) = \frac{1}{4}x^3 + 2$, what is the value of $f^{-1}(4)$?

- A. 2
- B. 5
- C. 8
- D. 18

Question 3 (1 mark)

In how many ways can 7 boys be arranged in a line if Peter, Jeffery, and Kevin have to stand together at the beginning of the line?

- A. $4!$
- B. $7!$
- C. $4! \times 3!$
- D. $7! \times 3!$

Question 4 (1 mark)

Which expression is equivalent to $\frac{1 + \tan 5x \tan 3x}{\tan 5x - \tan 3x}$

A. $\cot 2x$

B. $\tan 2x$

C. $\frac{\tan 2x}{1 + \tan 5x \tan 3x}$

D. $\cot x$

Question 5 (1 mark)

The roots of the equation $2x^3 - 5x^2 + 8x + 2 = 0$ are α , β and γ . Find the value of $\alpha(\beta^2 + \gamma^2) + \beta(\alpha^2 + \gamma^2) + \gamma(\alpha^2 + \beta^2)$

A. -4

B. $\frac{13}{2}$

C. 7

D. 13

Exam continues on the next page

Mathematics Extension 1

Section II

85 Marks

Attempt Questions 6-11

Start each question on a new page

Allow about 110 minutes for this section

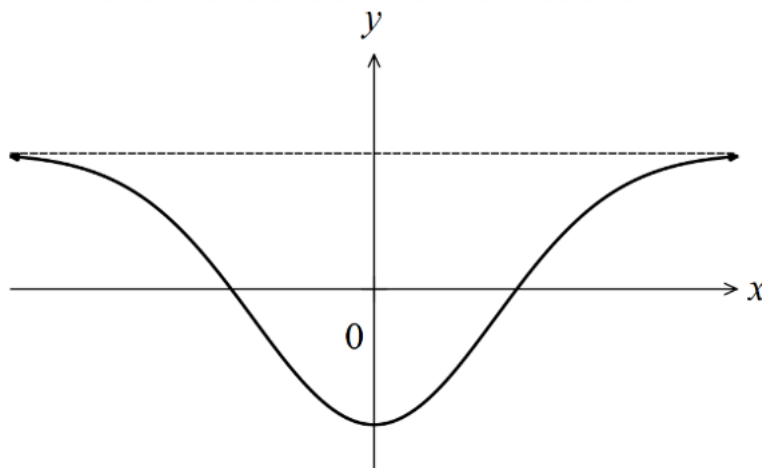
Question 6 (13 marks)

(a) Solve the inequality $\frac{x}{2x-1} \leq 5$ [3]

(b) Find the cartesian equation of the curve defined by the parametric equations: [2]

$$\begin{aligned}x &= 2 \sin \theta \\y &= 3 + 2 \cos 2\theta\end{aligned}$$

(c) The diagram below shows the graph of $y = f(x)$ where $f(x) = 1 - 2e^{-x^2}$.

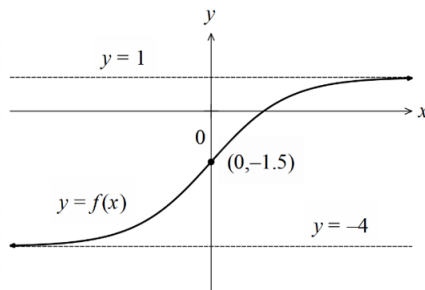


i. Find the equation of the horizontal asymptote, showing relevant working out. [1]

ii. A copy of the graph $y = f(x)$ is on page 4 of your writing booklet. On the same set of axes, sketch the graph of $y = \sqrt{f(x)}$. Showing clearly the asymptote and any points of intersection between $y = f(x)$ and $y = \sqrt{f(x)}$. [2]

Exam continues on the next page

- (d) The graph of the function $f(x) = \frac{e^x - 4}{e^x + 1}$ is shown below.



- i. Explain why $f(x)$ has an inverse function for all values of x . [1]
- ii. On the same set of axes which is on the page 5 of your solution booklet, sketch the graphs of $y = x$, $y = f(x)$ and $y = f^{-1}(x)$ [2]
- iii. Show that the equation of the inverse function is $f^{-1}(x) = \ln\left(\frac{x+4}{1-x}\right)$ [2]

Question 7 (12 marks)

- (a) What is the remainder when $P(x) = 2x^4 - 5x^3 + 12x - 7$ is divided by $x - 2$? [1]
- (b) Consider the polynomial $p(x) = x^3 - px^2 + qx - pq$, where $q > 0$.
 - i. Show that $(x - p)$ is a factor of $P(x)$. [1]
 - ii. Show that the equation $P(x) = 0$ has only one real root. [2]
- (c) Find the multiplicity of the root 1 of the equation $P(x) = 0$ where $P(x) = x^4 - 5x^3 + 9x^2 - 7x + 2$ [2]
- (d) Consider the polynomial $P(x) = 2x^3 - 19x^2 + 60x - 63$. It is given that $P(x)$ has a double root at $x = 3$, factorise $P(x)$ completely. [2]
- (e) The zeros of $x^3 - 5x + 3 = 0$ are α , β and γ , find a cubic equation with integer coefficients whose zeros are 2α , 2β and 2γ . [4]

Exam continues on the next page

Question 8 (14 marks)

- (a) Write $\frac{1}{(n+1)!} - \frac{1}{(n-1)!}$ as a single fraction [2]
- (b) If each distinct arrangement of the letters of DELETED is called a word:
- i. How many words are possible? [1]
 - ii. In how many of these words will the D's be separated? [1]
- (c) A five-person team is to be chosen at random from eight women and nine men.
- i. In how many ways can this team be chosen? [1]
 - ii. What is the probability that the team will consist of five men? [1]
- (d) Find the number of ways in which the letters of the word HEXAGON can be arranged in a straight line so that: [2]
- i. the vowels are all next to each other. [1]
 - ii. no two consonants are next to each other [1]
- (e) In how many ways can a committee of 5 be selected from 10 people, consisting of 5 men and 5 women, if one man, Ken, refuses to work in the same committee with Sue? [2]
- (f) Consider the digits 7, 6, 5, 4, 3, 2, 1 and 0. Find how many four-digit numbers are possible if the digits are in:
- i. descending order [1]
 - ii. ascending order [1]

Exam continues on the next page

Question 9 (13 marks)

(a) Using the expansion of $\cos(A + B)$, find the exact value of $\sec 105^\circ$ [3]

(b) i. Prove that $\tan \theta = \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$ [2]

ii. Hence show the exact value of $\tan 22\frac{1}{2}^\circ = \sqrt{2} - 1$ [2]

(c) By making the substitution $t = \tan \frac{\theta}{2}$. [2]

Show that $\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}$

(d) Show that $\sin(A - 2B) \cos(A + 2B) = \frac{1}{2}(\sin 2A - \sin 4B)$ [2]

Note: Sine is an odd function and Cosine is an even function.

(e) Prove the identity $\frac{2\sin^3 A + 2\cos^3 A}{\sin A + \cos A} = 2 - \sin 2A$ [2]

Exam continues on the next page

Question 10 (17 marks)

- (a) There are four couples: [2]

Aaryan and Betty
Chen and Daisy
Elias and Fiona
Gus and Helena

They go to a cinema and decide to sit in the last row. How many different arrangements are possible if Aaryan and Betty want to sit together but Elias and Fiona do NOT want to sit together?

- (b) Ken wants to celebrate his 18th birthday by having a dinner party for himself and nine of his friends (five girls and four boys).

In how many ways can the people be seated at a round table if:

- i. the boys and girls alternate? [1]

- ii. Ken to be seated between two particular girls? [1]

- (c) In how many ways 8 different gifts can be distributed to two children so that each child receives an odd number of gifts? [2]

- (d) Five points are scattered over a 2m by 2m square. Show that there is at least one pair of points which are a distance at most $\sqrt{2}$ m apart. [2]

- (e) Ten members of a sporting team have the numbers 1 to 10 on the back of their jumpers. What is the minimum number of players that must be chosen to guarantee that there is at least one pair of players whose numbers add to 11? [1]

- (f) i. Expand and simplify $(3 + x^2)^5$. [2]

- ii. Hence find the coefficient of x^4 in $(x + 2)^2(3 + x^2)^5$ [3]

- (g) Use binomial theorem to find an expression for the constant term in the expansion of $\left(3x^2 - \frac{1}{x}\right)^{15}$. Hence find the constant term. [3]

Exam continues on the next page

Question 11 (16 marks)

(a) By using $(1+x)^{10} = (1+x)^5(1+x)^5$, show that [2]

$$\binom{10}{3} = \binom{5}{3} \binom{5}{0} + \binom{5}{2} \binom{5}{1} + \binom{5}{1} \binom{5}{2} + \binom{5}{0} \binom{5}{3}$$

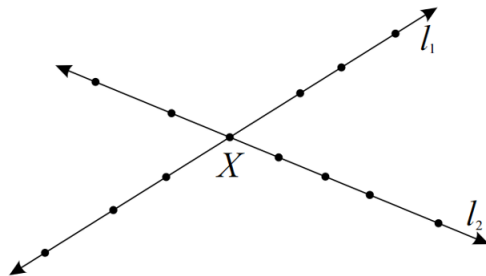
(b) In the binomial expansion of $\left(1 + \frac{x}{k}\right)^n$, the coefficient of x^3 is twice the coefficient of x^2 . Prove that $n = 6k + 2$ [2]

(c) i. Write down the binomial expansion of $(1+x)^n$. [1]

ii. Hence, show that $\binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \dots + \binom{n}{n} = 2^n - 1$. [2]

iii. By substituting a suitable value for x in the expansion $(1+x)^n$, show that: [2]
 $1 + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^{n-1}\binom{n}{n-1} = 3^n - 2^n$

Exam continues on the next page



- (d) The diagram above shows two distinct intersecting lines l_1 and l_2 lying on a plane. Each line has n distinct points marked on it, with the point X lying on both lines. The case above shows a possible situation when $n = 7$, and there are a total of thirteen points marked.

Consider the general case where there are n distinct points marked on each line, for $n \geq 3$.

- i. In how many ways can three points be chosen on l_1 ? [1]
- ii. By considering the total number of points on both lines and your answer to part(i), determine the number of triangles that can be formed using the marked points on either lines as vertices. [2]
- iii. Of the total number of possible triangles in part (ii):
 - α) how many have X as a vertex? [1]
 - β) how many have two vertices that lie on l_1 ? [1]
- iv. Use your answers to part (ii) and part (iii) to show that: [2]

$$\binom{2n-1}{3} = 2 \times \binom{n}{3} + (n-1)^3.$$

End of Examination

Mathematics Assessment Solutions
Girraween High School
2024

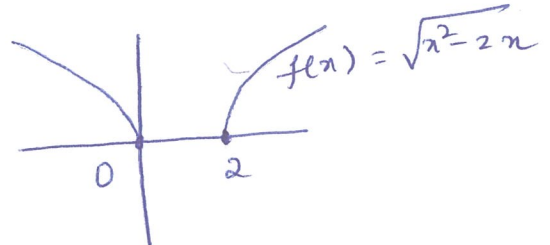
Task: Y11 Yearly Exam

Year: 2024

Suggested Solutions

Question 1

(C) $x \leq 0$ or $x > 2$



Question 2

(A)

$$y = \frac{1}{4}x^3 + 2$$

$$x = \frac{1}{4}y^3 + 2$$

$$y^3 = 4x - 8$$

$$y = \sqrt[3]{4x - 8}$$

$$f^{-1}(4) = \sqrt[3]{4(4) - 8}$$

$$= 2$$

1	2	3	4	5
C	A	C	A	D

Marker's Feedback

Suggested Solutions

Question 3

(C) $4! \times 3!$

Question 4 (A)

$$\frac{1 + \tan 5x \tan 3x}{\tan 5x - \tan 3x}$$

$$= \frac{1}{\frac{\tan 5x - \tan 3x}{1 + \tan 5x \tan 3x}}$$

$$= \frac{1}{\tan(5x - 3x)}$$

$$= \frac{1}{\tan 2x}$$

$$= \cot 2x$$

Marker's Feedback

Suggested Solutions

Question 5

(D)

$$\alpha + \beta + \gamma = \frac{5}{2}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = 4$$

$$\alpha\beta\gamma = -1$$

$$\alpha\beta^2 + \alpha\gamma^2 + \beta\alpha^2 + \beta\gamma^2 + \gamma\alpha^2 + \gamma\beta^2$$

$$\alpha\beta^2 + \beta\alpha^2 + \alpha\gamma^2 + \gamma\alpha^2 + \beta\gamma^2 + \gamma\beta^2$$

$$\alpha\beta(\alpha + \beta) + \alpha\gamma(\alpha + \gamma) + \beta\gamma(\gamma + \beta)$$

$$\alpha\beta(\alpha + \beta + \gamma) - \alpha\beta\gamma + \alpha\gamma(\alpha + \beta + \gamma) - \alpha\beta\gamma + \beta\gamma(\alpha + \beta + \gamma) - \alpha\beta\gamma$$

$$(\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - 3(\alpha\beta\gamma)$$

$$4\left(\frac{5}{2}\right) - 3(-1)$$

$$= 13$$

Marker's Feedback

Suggested Solutions

Question 6

$$\textcircled{a} \quad \frac{x}{2x-1} \leq 5 \quad x \neq \frac{1}{2}$$

$$(2x-1)^2 \times \frac{x}{(2x-1)} \leq 5(2x-1)^2$$

$$x(2x-1) \leq 5(2x-1)^2$$

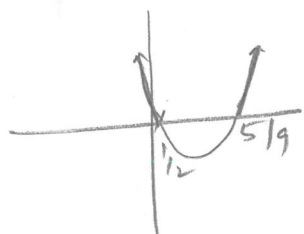
— $\textcircled{1}$

$$0 \leq 5(2x-1)^2 - x(2x-1)$$

$$0 \leq (2x-1) [5(2x-1) - x]$$

$$0 \leq (2x-1)(9x-5)$$

$$(2x-1)(9x-5) \geq 0$$

— $\textcircled{1}$

$$x < \frac{1}{2} \text{ and } x > \frac{5}{9} \quad \& \; x \neq \frac{1}{2} \quad \text{---} \quad \textcircled{1}$$

Marker's Feedback

Some students who did not show the diagram as part of the above shown working ~~but~~ have lost a mark

→ some students forgot to omit $x = \frac{1}{2}$ as part of the solution,

Suggested Solutions

Question 6 b

$$x = 2 \sin \theta$$

$$x^2 = 4 \sin^2 \theta$$

→ (1)

$$y = 3 + \cos 2\theta$$

$$y = 3 + 1 - 2 \sin^2 \theta$$

$$y = 4 - 2 \sin^2 \theta$$

→ (2)

From (1) and (2)

$$\sin^2 \theta = \frac{x^2}{4} \text{ sub into eq (2)}$$

— (1M)

$$y = 4 - 2 \left(\frac{x^2}{4} \right)$$

$$y = 4 - \frac{x^2}{2}$$

$$\therefore 2y = 8 - x^2$$

— (1M)

Marker's Feedback

Students who did not realise the double angle or couldn't use the double angle formula have lost marks -

Suggested Solutions

Question 6C

i, let $y = f(x)$

$$y = 1 - \frac{2}{e^{x^2}}$$

$$\text{As } x \rightarrow \infty \quad \frac{-2}{e^{x^2}} \rightarrow 0^-$$

$$1 - \frac{2}{e^{x^2}} \rightarrow 1^-$$

$$\text{As } x \rightarrow -\infty \quad \frac{-2}{e^{x^2}} \rightarrow 0^-$$

$$1 - \frac{2}{e^{x^2}} \rightarrow 1^-$$

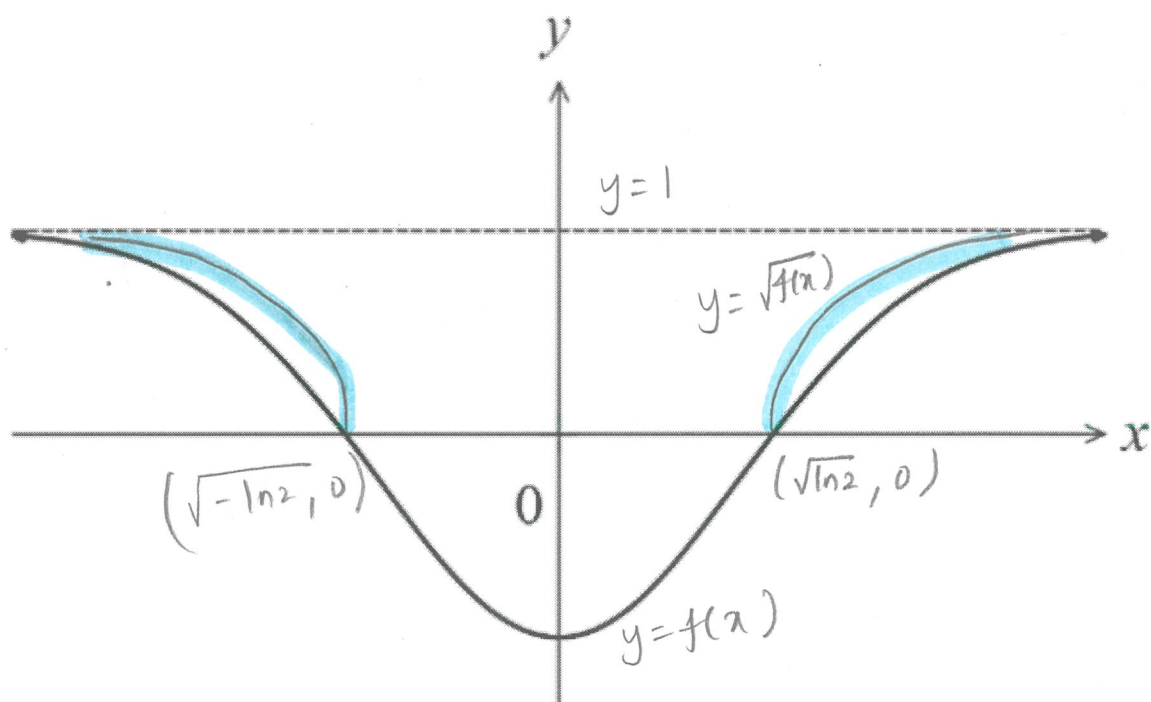
$\therefore$ Horizontal asymptote is $y = 1$

(1M)

Marker's Feedback

Mostly done well; however some students did not write the equation $y = 1$; instead wrote Horizontal asymptote = 1; It is recommended to always write the equation.

Q6Cii



Correct shape or position (1m)

Correct shape or position relative to $f(x)$ $\rightarrow$ (2m)

Mostly done well otherwise lost marks.

Suggested Solutions

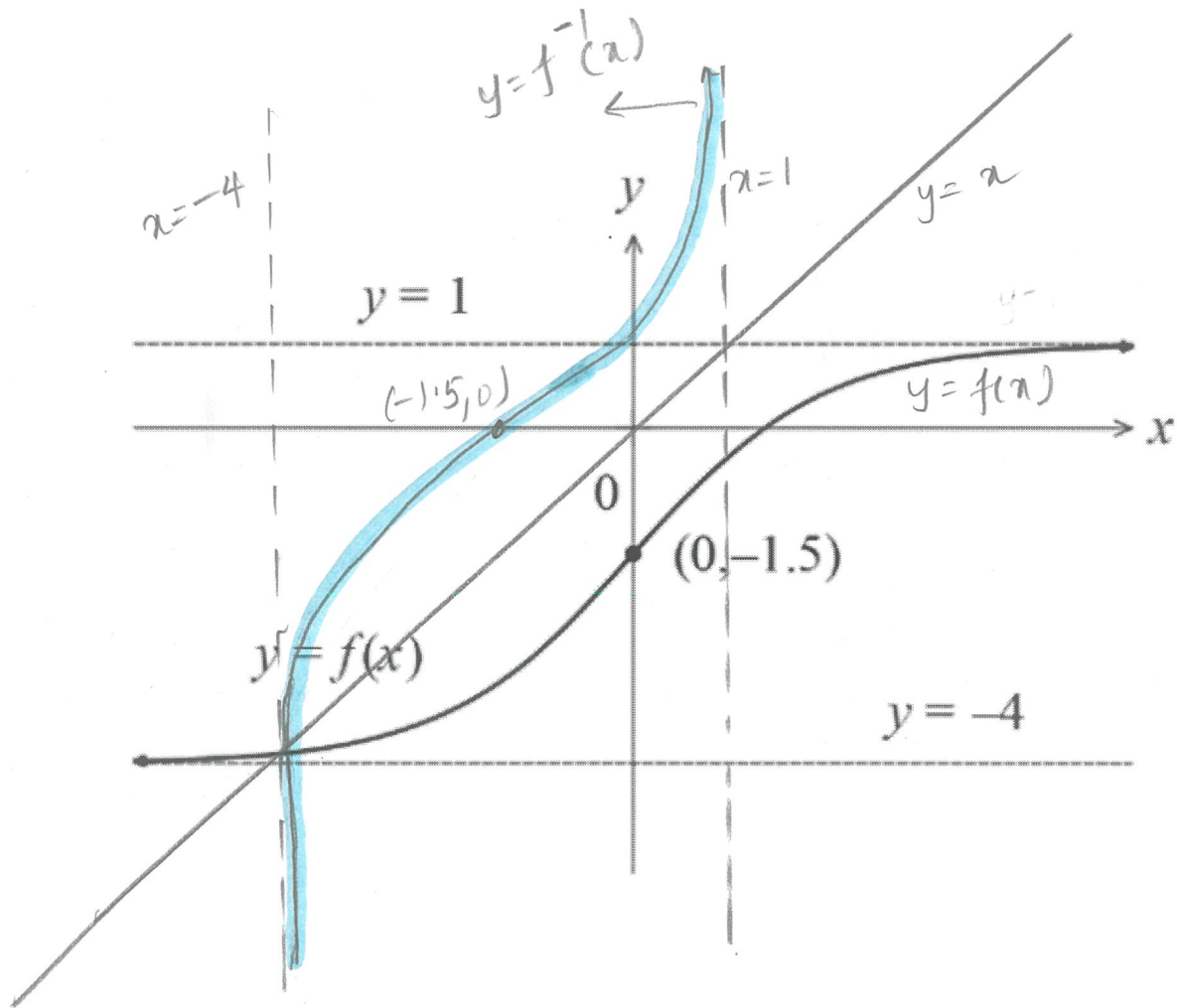
Question 6d

i, $f(x)$ is a one-one-one function or a monotonically increasing function.

Marker's Feedback

Mostly done well; Most students either wrote one-one-one function or passed Horizontal line test and Vertical line test.

Q6 d, ii,



Few students lost all marks.
(Students advised to revise similar style question)

Suggested Solutions

Question 6d(iii),

$$f(x) = y = \frac{e^x - 4}{e^x + 1}$$

$$f^{-1}(x) = x = \frac{e^y - 4}{e^y + 1}$$

— (1m)

$$xe^y + x = e^y - 4$$

$$(x-1)e^y = -4-x$$

$$e^y = \frac{x+4}{1-x}$$

$$y = \ln\left(\frac{x+4}{1-x}\right); \text{ as required } \rightarrow (1m)$$

Marker's Feedback

Mostly done well!

Suggested Solutions

Question 7a

$$p(2) = 2(2)^4 - 5(2)^3 + 12(2) - 7$$

$$= 9 \quad - (1m)$$

Well done.
Some students
used division
which is not efficient.

(7b) i, $p(x) = x^3 - px^2 + qx - pq$; $q > 0$

if $(x-p)$ is a factor then
 $p(p)$ will be equal to zero.

$$p(p) = \cancel{p^3} - p(\cancel{p})^2 + q(\cancel{p}) - \cancel{p}q$$

$$= 0$$

$\therefore (x-p)$ is a factor

well done
- (1)

(7b ii) $p(x) = x^3 - px^2 + qx - pq$

$$= x^2(x-p) + q(x-p)^2$$

$$= (x^2 + q)(x-p)$$

Students had
trouble with
this.

- (1)

Now $x^2 + q = 0$ has no real roots

$\therefore p(x) = 0$ has only one real root i.e. p .

- (1) must
provide
some sort
of reasoning
to get second
mark.

Marker's Feedback

Question

7C

$$p(x) = x^4 - 5x^3 + 9x^2 - 7x + 2$$

$$; p(1) = 0$$

$$p'(x) = 4x^3 - 15x^2 + 18x - 7$$

$$; p'(1) = 0$$

$$p''(x) = 12x^2 - 30x + 18$$

$$; p''(1) = 0$$

$$p'''(x) = 24x - 30$$

$$; p'''(1) \neq 0 \rightarrow \textcircled{1}$$

$\therefore 1$ is a root of multiplicity 3.

generally
well done

7d) consider the polynomial

$$p(x) = 2x^3 - 19x^2 + 60x - 63$$

$$; p(3) = 0$$

Given that $x=3$ is a double root,

$$\therefore p(x) = (x-3)^2(ax+b)$$

— (1M)

$$= (x^2 - 6x + 9)(ax+b)$$

Comparing the coefficients of x^3 : $a=2$

comparing the constant term, $b=-7$

$$\therefore p(x) = (x-3)^2(2x-7)$$

— (1M)
well done.

Marker's Feedback

Alternate solution for
7d let the roots be

$$\alpha, 3, 3$$

$$\therefore \alpha + 3 + 3 = \frac{9}{2}$$

$$\therefore \alpha = \frac{1}{2}$$

$$p(x) = (x-3)^2(2x-7)$$

A common error was
 $p(x) = (x-3)^2(x-\frac{1}{2})$

Question (7e)

$$\begin{aligned} \alpha + \beta + \gamma &= -\frac{b}{a} = 0 \\ 2\alpha + 2\beta + 2\gamma &= 2(\alpha + \beta + \gamma) \\ &= 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} \alpha + \beta + \gamma &= -\frac{b}{a} = 0 \\ 2\alpha + 2\beta + 2\gamma &= 2(\alpha + \beta + \gamma) \\ &= 0 \end{aligned}} \right\} (1M)$$

$$\begin{aligned} \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} = -5 \\ 2\alpha \cdot 2\beta + 2\alpha \cdot 2\gamma + 2\beta \cdot 2\gamma &= 4(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= -20 \end{aligned} \quad \left. \vphantom{\begin{aligned} \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} = -5 \\ 2\alpha \cdot 2\beta + 2\alpha \cdot 2\gamma + 2\beta \cdot 2\gamma &= 4(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= -20 \end{aligned}} \right\} (1M)$$

$$\begin{aligned} \alpha\beta\gamma &= -\frac{d}{a} = -3 \\ 2\alpha \cdot 2\beta \cdot 2\gamma &= 8\alpha\beta\gamma \\ &= -24 \end{aligned} \quad \left. \vphantom{\begin{aligned} \alpha\beta\gamma &= -\frac{d}{a} = -3 \\ 2\alpha \cdot 2\beta \cdot 2\gamma &= 8\alpha\beta\gamma \\ &= -24 \end{aligned}} \right\} (1M)$$

$\therefore$ New cubic equation with integer coefficients
is $x^3 - 20x + 24 = 0$ (1M)

was not too poorly done.
common errors $2\alpha \cdot 2\beta \cdot 2\gamma = 2\alpha\beta\gamma$
 $2\alpha\beta + 2\beta\gamma + 2\alpha\gamma = 2(\dots)$

Marker's Feedback

Suggested Solutions

Question 8a

$$\frac{1}{(n+1)!} - \frac{1}{(n-1)!}$$

$$\frac{1}{(n-1)!} \left[\frac{1}{n(n+1)} - 1 \right] \quad - (1M)$$

$$\frac{1}{(n-1)!} \left[\frac{1 - (n(n+1))}{n(n+1)} \right]$$

$$\frac{1}{(n-1)!} \left[\frac{1 - n^2 - n}{n(n+1)} \right]$$

$$= \frac{1 - n^2 - n}{(n+1)!} \quad - (1M)$$

E

Marker's Feedback

Generally well done, although many students stopped at $\frac{1 - n(n+1)}{(n+1)!} \rightarrow$ note: this was still paid full marks.

overexpanded this to $\frac{1 - n^2 - n}{(n+1)!}$

Suggested Solutions

Question 8b.i,

$$\begin{aligned} \text{N.o of ways} &= \frac{7!}{2!3!} \\ &= 420 \quad \text{--- (1m)} \end{aligned}$$

$$(8b.ii) \quad \text{All D's together} = \frac{6!}{3!} = 120 \text{ ways}$$

$$\text{D's separated is } 420 - 120 = 300 \text{ ways}$$

→ (1m)

$$(8c.i) \quad {}^{17}C_5 = 6188$$

$$(8c.ii) \quad \frac{{}^9C_5}{{}^{17}C_5} = \frac{126}{6188} = \frac{9}{442}$$

Marker's Feedback

(8)(b)&(c) Generally well done by most students.

The few students who didn't get these right generally didn't understand at all.

Suggested Solutions

Question

8d.i,

Vowels : $\boxed{E, A, O}$ & H & G & N can be arranged in $5!$ ways

$\boxed{E, A, O}$ can be arranged in $3!$ ways

$$\therefore 5! \times 3! = 720 \text{ ways.}$$

8d.ii No consonants are next to each other
V - Vowels ; C - Consonants

can only be arranged as

C V C V C V C

C arrangements = $4!$

V arrangements = $3!$

$$\therefore 4! \times 3! = 144 \text{ ways.}$$

Marker's Feedback

Once again, generally well done with the few students who didn't understand being completely lost.

Suggested Solutions

Question

(8e)

K, Not Sue $8C_4$ waysNot Ken, Sue $8C_4$ waysNo Ken, No Sue $8C_5$ ways

} (1m)

 $\therefore$ Number of ways = 196

— (1m)

(8f) i, descending order

7, 6, 5, 4, 3, 2, 1 & 0

Selecting any four digits and then arranged in the required order: $8C_4 = 70$ (1m)

(8f) ii Ascending Order:

Cannot start with zero

 $\therefore 7C_4 = 35$ $\rightarrow$ (1m)

Marker's Feedback

Alternative solution was (for e)

Total — both together
= $10C_5 - 8C_3$
= 196.

Students did both this way & the above method well.

Also in (e); Some students had the right idea but put $9C_4$ for Ken not Sue (forgetting that BOTH Ken & Sue are accounted for so there are only 8 people left).

(f) Generally well done. Most students recognised that a COMBINATION could be used & that you could not start with 0 for ascending order.

The students who didn't get this generally had no idea (again).

Suggested Solutions

Question

9a

$$\sec 105^\circ = \frac{1}{\cos(105^\circ)} = \frac{1}{\cos(60^\circ + 45^\circ)} \quad (1M)$$

$$= \frac{1}{\cos 60 \cos 45 - \sin 60 \sin 45} \quad (1M)$$

$$= \frac{1}{\left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right) - \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right)}$$

$$= \frac{2\sqrt{2}}{1 - \sqrt{3}} \quad \text{or} \quad (1M)$$

$$\frac{2\sqrt{2}(1 + \sqrt{3})}{1 - 3}$$

$$= -\sqrt{2} - \sqrt{6}$$

Marker's Feedback

Well done.

Suggested Solutions

Question

(9b i)

$$\tan \theta = \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$$

$$\text{RHS} = \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$$

$$\text{RHS} = \sqrt{\frac{1 - (1 - 2\sin^2 \theta)}{1 + (2\cos^2 \theta - 1)}}$$

$$= \sqrt{\frac{1 - 1 + 2\sin^2 \theta}{1 + 2\cos^2 \theta - 1}}$$

$$= \sqrt{\frac{2\sin^2 \theta}{2\cos^2 \theta}}$$

$$= \tan \theta ; \text{ as required}$$

— (1m)

(1m)

Marker's Feedback

Mostly done well.

Suggested Solutions

Question

9bii,

$$\tan 22\frac{1}{2} = \sqrt{\frac{1 - \cos(22\frac{1}{2} \times 2)}{1 + \cos(22\frac{1}{2} \times 2)}}$$

$$= \sqrt{\frac{1 - \cos 45}{1 + \cos 45}}$$

(1M)

$$= \sqrt{\frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}}}$$

$$= \sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1} \times \frac{(\sqrt{2}-1)}{(\sqrt{2}-1)}}$$

$$= \sqrt{\frac{(\sqrt{2}-1)^2}{2-1}}$$

$$= \sqrt{(\sqrt{2}-1)^2}$$

$$= \sqrt{2} - 1$$

(1M)

Marker's Feedback

Some students didn't show that $\tan 22\frac{1}{2} = \sqrt{2} - 1$
 Some got $\tan 22\frac{1}{2} = \sqrt{3-2\sqrt{2}}$ or $\sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1}}$
 This is a "show that" question, so '1' mark ^{out} deducted if they didn't show full working to get the final answer $\sqrt{2} - 1$.

Suggested Solutions

Question

(9C) $t = \tan \frac{\theta}{2}$

Show that $\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}$.

$$\text{LHS: } \frac{1+t^2}{2t} + \frac{1-t^2}{2t}$$

$$= \frac{2}{2t}$$

— ①

$$= \frac{1}{t}$$

$$= \frac{1}{\tan \frac{\theta}{2}}$$

— ①

$$= \cot \frac{\theta}{2}$$

Marker's Feedback

Done well.

Suggested Solutions

Question

(9d)

$$\text{LHS: } \sin(A-2B) \cos(A+2B)$$

→ (1M)

$$= \frac{1}{2} [\sin(A-2B+A+2B) + \sin(A-2B-A-2B)]$$

$$= \frac{1}{2} [\sin 2A + \sin(-4B)] \rightarrow (1M)$$

$$= \frac{1}{2} [\sin 2A - \sin 4B] \quad (\text{since is odd})$$

↓
no mark assigned.

$$= \text{RHS; as required.}$$

Marker's Feedback

Generally well done.

Suggested Solutions

Question

(9c)

$$\text{LHS: } \frac{2\sin^3 A + 2\cos^3 A}{\sin A + \cos A}$$

$$\frac{2(\sin^3 A + \cos^3 A)}{\sin A + \cos A}$$

$$= \frac{2(\sin A + \cancel{\cos A})(\sin^2 A - \sin A \cos A + \cos^2 A)}{(\sin A + \cancel{\cos A})} \quad (1M)$$

$$= 2(\sin^2 A + \cos^2 A - \sin A \cos A)$$

$$= 2(1 - \sin A \cos A) \rightarrow (1M)$$

$$= 2 - 2\sin A \cos A$$

$$= 2 - \sin 2A$$

$$= \text{RHS}$$

Marker's Feedback

Some students had trouble with this question.

Question 10a

AB

C D

E F

G H

Separate

working out
→ 1mAll arrangements with AB — AB EF arrangements
enti

$$7! \times 2 - 6! \times 2! \times 2!$$

$$= 1200 \quad \text{---} \quad 1m$$

10b i

place one boy first, the rest can
be arranged in $4! \times 5! = 2880$ ways.

→ 1m

10b ii

Place Ken and the two girls first
This can be done in $2!$ waysThe remainder can be arranged in
 $7!$ ways

$$\therefore \text{N.o of ways} = 2! \times 7! = 10080 \text{ ways}$$

→ 1m

Marker's Feedback

Most of the students did well but some
students got all Perms and combs questions
wrong.

Suggested Solutions

Question

10C

A 1 3 5 7

B 7 5 3 1

→ (1m)

N.O of ways: ${}^8C_1 \cdot {}^7C_7 + {}^8C_3 \cdot {}^5C_5 + {}^8C_5 \cdot {}^3C_3 + {}^8C_7 \cdot {}^1C_1$

= 128 ways → (1m)

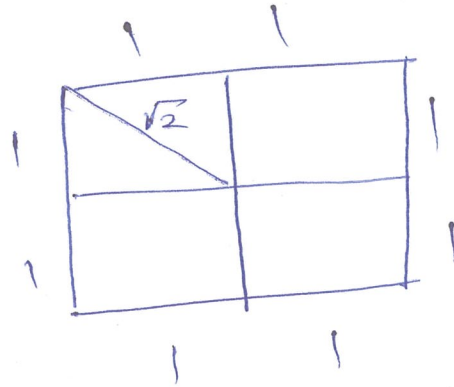
Marker's Feedback

Lot of Students gave 4 as the answer

(10d)

Divide the square into $1m \times 1m$ squares
By the pigeonhole principle, at least one of the squares must contain 2 points. The furthest distance between the 2 points is the diagonal of the smaller square which is $\sqrt{2}m$.

(2m)



Marker's Feedback

Majority got it right - well done.

$$\{1, 10\}, \{2, 9\}, \{3, 8\}, \{4, 7\}, \{5, 6\}$$

worst case scenario

Choosing 5 players whose numbers do not add to 11.

But choosing 1 more would guarantee that there is at least one pair of players whose numbers add to 11.

∴ 6 is the minimum number of players required.

↳ (1m)

Mostly done well.

Suggested Solutions

Question

109 $\left(3x^2 - \frac{1}{x}\right)^{15}$

General term: ${}^{15}C_k (3x^2)^{15-k} \left(-\frac{1}{x}\right)^k$

$${}^{15}C_k 3^{15-k} x^{30-2k} (-1)^k x^{-k}$$

$${}^{15}C_k 3^{15-k} x^{30-3k} (-1)^k$$

(1m)

Constant term occurs when $30 - 3k = 0$

$$30 = 3k$$

$$k = 10$$

→ (1m)

$$\therefore \text{constant term} = {}^{15}C_{10} \times 3^{15-10} \times (-1)^{10}$$

$$= 729729$$

→ (1m)

Marker's Feedback

Mostly done well.

Some had -729729 as answer
and lost 1 mark.

Suggested Solutions

Question

(10f.i)

$$\begin{aligned}
 (3+x^2)^5 &= {}^5C_0 \cdot 3^5 + {}^5C_1 \cdot 3^4 \cdot x^2 + {}^5C_2 \cdot 3^3 \cdot (x^2)^2 + {}^5C_3 \cdot 3^2 \cdot (x^2)^3 + \\
 & {}^5C_4 \cdot 3 \cdot (x^2)^4 + {}^5C_5 \cdot (x^2)^5 \quad \rightarrow (1m)
 \end{aligned}$$

$$= 243 + 405x^2 + 270x^4 + 90x^6 + 15x^8 + x^{10} \quad \rightarrow (1m)$$

Coefficient of x^4

(10f.ii)

$$(x+2)^2 (3+x^2)^5$$

$$(x^2+4x+4) (3+x^2)^5 \quad \rightarrow (1m) \quad \text{using (9.i.i.)}$$

$$405x^2 \cdot x^2 + 4 \times 270 \cdot x^4 \quad \rightarrow (1m)$$

$$\text{Coefficient of } x^4 \text{ is } \underline{1485}.$$

$$\rightarrow (1m)$$

Marker's Feedback

Majority did well.

10f.ii) Some put answer as $1485x^4$ rather than 1485.

Suggested Solutions

Question

(11a)

$$(1+x)^{10} = (1+x)^5 (1+x)^5$$

$$\text{LHS } (1+x)^{10} = {}^{10}C_0 + {}^{10}C_1 x + {}^{10}C_2 x^2 + {}^{10}C_3 x^3 + \dots$$

$$\text{coefficient of } x^3 = {}^{10}C_3 \rightarrow (1\text{m})$$

$$\text{RHS: } (1+x)^5 (1+x)^5 = \left({}^5C_0 + {}^5C_1 x + {}^5C_2 x^2 + {}^5C_3 x^3 + {}^5C_4 x^4 + {}^5C_5 x^5 \right) \left({}^5C_0 + {}^5C_1 x + {}^5C_2 x^2 + {}^5C_3 x^3 + {}^5C_4 x^4 + {}^5C_5 x^5 \right)$$

$$\text{coefficient of } x^3 = {}^5C_0 \cdot {}^5C_3 + {}^5C_1 \cdot {}^5C_2 + {}^5C_2 \cdot {}^5C_1 + {}^5C_3 \cdot {}^5C_0 \rightarrow (1\text{m})$$

Equating coefficients of x^3

$${}^{10}C_3 = {}^5C_3 \cdot {}^5C_0 + {}^5C_2 \cdot {}^5C_1 + {}^5C_1 \cdot {}^5C_2 + {}^5C_0 \cdot {}^5C_3$$

$${}^{10}C_3 = 2 \left[{}^5C_0 \cdot {}^5C_3 + {}^5C_1 \cdot {}^5C_2 \right]$$

Marker's Feedback

Some students didn't expand the LHS and RHS. Without expanding and writing you can't equate coefficients. They lost 1 mark.

Suggested Solutions

Question

(11b)

$$2 \cdot {}^nC_2 \frac{x^2}{k^2} \quad ; \quad {}^nC_3 \frac{x^3}{k^3}$$

$$\frac{2n!}{(n-2)! 2! k^2} = \frac{n!}{(n-3)! 3! k^3}$$

$$\frac{\cancel{2} n (\cancel{n-1}) (\cancel{n-2})!}{(\cancel{n-2})! \cancel{2!} k^2} = \frac{n (n-1) (n-2) (\cancel{n-3})!}{(\cancel{n-3})! 3! \cancel{k^3} k}$$

$$n(n-1) = \frac{n(n-1)(n-2)}{3! k}$$

$$6k = n-2$$

$$\therefore n = 6k + 2$$

Marker's Feedback

Working out very messy. students do not know the difference between term and coefficient.

Suggested Solutions

Question

(11C)

$$i, (1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$$

$$ii, \text{ Sub } x=1 \text{ in } i, \quad (1M)$$

$$2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n$$

$$2^n = 1 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n \rightarrow (1M)$$

$$\therefore {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n - 1$$

$$iii, \text{ Sub } x=2 \text{ in } i, \quad (1M)$$

$$3^n = {}^nC_0 + 2 \cdot {}^nC_1 + 4 \cdot {}^nC_2 + \dots + 2^{n-1} \cdot {}^nC_{n-1} + 2^n \cdot {}^nC_n$$

$$3^n = \underline{{}^nC_0} + 2 \cdot {}^nC_1 + \dots + 2^{n-1} \cdot {}^nC_{n-1} + \underline{2^n} \quad (1M)$$

$$\therefore 3^n - 2^n = 1 + 2 \binom{n}{1} + 4 \binom{n}{2} + \dots + 2^{n-1} \binom{n}{n-1};$$

as required

Marker's Feedback

(i) done well

(ii) done well

(iii) Some students subtracted the value of 2^n from (ii) and lost marks.

11d

(i) nC_3 ways $\rightarrow 1m$

(ii) There are $2n-1$ points in total, and any 3 non-collinear points can be chosen to form the vertices of a Triangle.

$\therefore$ there are ${}^{2n-1}C_3 - 2 \times {}^nC_3$ possible triangles.

(iii), (d) If 'x' is assigned as a vertex:

$\rightarrow$ 1 vertex must be chosen from l_1 which happens in $(n-1)$ ways.

$\rightarrow$ 1 vertex must be chosen from l_2 $(n-1)$ ways.

$\therefore (n-1)^2$ triangles have x as vertex.

Marker's Feedback

(i) Some students wrote nC_3 . They were not given mark.

(ii) Hard question. very few got correct answer.

(iii) Hard question. Not many correct answers.

Question 11d (continued)

(iii) $\beta \rightarrow n_{C_2}$ ways to choose the vertices on l_1
 $\rightarrow (n-1)$ ways to choose the remaining vertex on l_2
 $n_{C_2} \times (n-1)_{C_1}$ triangles.

(iv) The total number of triangles that can be formed can be considered as:

$$\left(\text{Triangles with 2 vertices on } l_1 \right) + \left(\text{Triangles with 2 vertices on } l_2 \right) - \left(\text{Triangles that have X as vertex, which are counted twice} \right)$$

$\therefore$ From (iii), Total Δ^{les}

$$\begin{aligned} &= n_{C_2}(n-1) + n_{C_2}(n-1) - (n-1)^2 \\ &= 2 \times \frac{n!}{2!(n-2)!} (n-1) - (n-1)^2 \\ &= n(n-1)^2 - (n-1)^2 \\ &= (n-1)^2(n-1) = (n-1)^3 \end{aligned}$$

Marker's Feedback

(iii) β hard question.
 (iv) Very hard question.

Suggested Solutions

Question

11d.iv, continued

Equating this result with result from (b) ii :

$$2n-1 \text{C}_3 - 2 \times n \text{C}_3 = (n-1)^3$$

$$2n-1 \text{C}_3 = 2 \times n \text{C}_3 + (n-1)^3 ; \text{ as required.}$$

Marker's Feedback

Only one or two got the correct answer.